

MMM Technical Architecture & Code Flow

This document describes **how the Auto-Modeller is implemented in code**.

1. System Overview

Pipeline:

Data → Validation → Tensor Builder → Bayesian Model → Sampling → Diagnostics → Contribution → ROI → Outputs

2. Main Execution Script

Entry point:

run_mmm_pipeline.py

This orchestrates the entire modelling process.

3. Pipeline Flow

```
load_data()
↓
validate_schema()
↓
validate_info()
↓
validate_data()
↓
cross_checks()
↓
convert_monthly_to_weekly()
↓
build_tensors()
↓
load_spend()
↓
run_stable_model()
↓
extract_contributions()
↓
compute_roi()
```

↓

compute_marginal_roi()

↓

write_outputs()

4. Code Structure

src/

io/

load_data.py

load_spend.py

write_outputs.py

validation/

validate_info.py

validate_data.py

cross_checks.py

parser/

build_tensors.py

preprocessing/

monthly_to_weekly.py

model/

pymc_mmm.py

contributions.py

roi.py

effectiveness.py

marginal_roi.py

roi_key_points.py

diagnostics/

diagnostics.py

5. Tensor Construction

Data is converted into tensors.

Shape:

(T, G, SB, C)

Where

dimension meaning

T time

G region

SB sub_brand

C channel

6. PyMC Model Builder

File:

pymc_mmm.py

Responsibilities:

- build probabilistic model
- define priors
- implement adstock
- implement saturation

7. Sampling Module

sample_model.py

Handles Bayesian sampling.

Includes:

- adaptive sampling
- divergence monitoring
- stability levels

8. Stability Controller

The system automatically adjusts model complexity.

Levels:

Level 0 – full flexibility

Level 1 – stronger priors

Level 2 – tighter constraints

This prevents model failure.

9. Contribution Extraction

File:

contributions.py

Uses **delta method**:

Prediction with channel

minus

Prediction without channel

10. ROI Calculation

File:

roi.py

Computes:

contribution_sales

ROI

11. Marginal ROI Calculation

File:

marginal_roi.py

Generates spend grid:

0.2x current spend

to

2x current spend

Evaluates incremental revenue.

12. Output Generator

File:

write_outputs.py

Outputs include:

coefficients.csv

roi.csv

media_contribution.csv

base_contribution.csv

predictions_timeseries.csv

marginal_roi_curves.csv

roi_key_points.csv

13. Runtime

Typical runtime:

model level	runtime
national	3-5 min
region	8-15 min
region × brand	15-30 min

14. Extensibility

The architecture allows future modules:

budget optimizer

scenario planner

forecasting

automated retraining